CS 486/686
Dissecting a Small Language Model

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Lecture 21

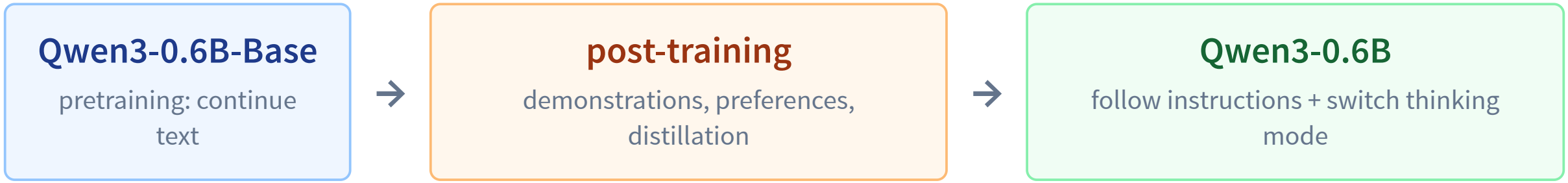
One real Qwen turn, end to end

Search › Uncertainty › Decisions › Learning

By the end, you can dissect one Qwen turn

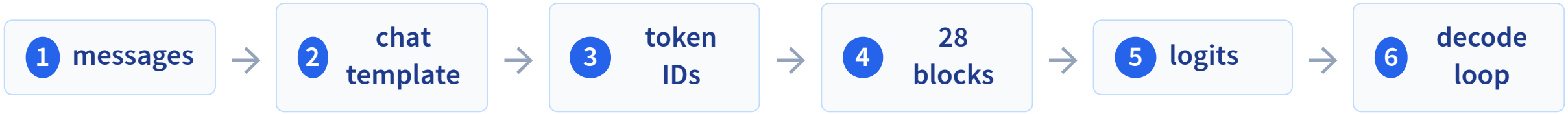
- Run **Qwen3-0.6B** in Python or a browser.
- Trace messages → template → tokens → hidden states → logits.
- Map real Qwen config fields onto a transformer block.
- Explain **prefill, decode, and the KV cache**.
- Predict how decoding and prompts change behavior.
- Recognize where a fixed model is insufficient.

L20 built a base model. Today we run an assistant.



Same causal architecture; new behavior learned after pretraining.

One user turn crosses six boundaries



We will inspect the real object at every boundary.

The real model we will dissect

Qwen3-0.6B

dense, decoder-only, post-trained

28 layers	**1024** hidden width	**3072** FFN width	**16 / 8** Q / KV heads
128 head width	**~152k** output scores	**32k** supported context	**tied** input/output matrix

Qwen Team, “Qwen3 Technical Report,” 2025.

Config is architecture written as data

```
{
  "hidden_size": 1024,
  "intermediate_size": 3072,
  "num_hidden_layers": 28,
  "num_attention_heads": 16,
  "num_key_value_heads": 8,
  "head_dim": 128,
  "vocab_size": 151936,
  "tie_word_embeddings": true
}
```

state	residual stream
communicate	16 query heads share 8 K/V heads
compute	3072-wide SwiGLU sublayer
predict	151,936 logits; input/output weights tied

The API receives structured messages

```
messages = [
    {
        "role": "user",
        "content": "Explain gradient descent in one senten
    }
]
```

role
who is speaking?

content
what did they say?

The model still consumes tokens—so this structure must become one string.

The chat template serializes one turn

```
<|im_start|>user
Explain gradient descent in one sentence.<|im_end|>
<|im_start|>assistant
generation starts here
```

No system message appears unless we actually supplied one.

Inspect the exact serialized input LIVE DATA

Raw messages → Qwen special tokens → thinking ON/OFF generation cue.

explanation | reasoning | future fact

thinking: OFF

structured message

user Explain gradient descent in one sentence.

exact string sent to tokenizer

```
<|im_start|>user
Explain gradient descent in one sentence.
<|im_end|>
<|im_start|>assistant
<think>

</think>
```

20 tokens after serialization

Template pre-fills an empty <think></think>; Qwen starts the answer.

Blue marks chat-control tokens. Green marks the thinking protocol. The user content is unchanged.

Tokenization happens after templating

<|im_start|>	user	↵	Ex	plain	·gradient	·descent	·in	·one
151644	872	198	840	20772	20169	36952	304	825

·sentence	.
11652	13

… role markers + generation cue

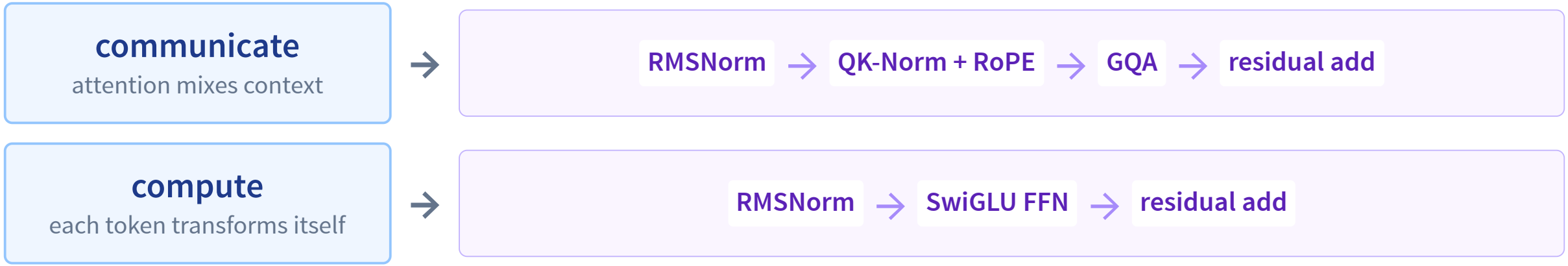
· means a leading space. IDs shown are actual Qwen tokenizer outputs, not illustrative.

Token IDs become a $[T, 1024]$ residual stream

Qwen position: RoPE rotates query/key dimensions inside attention; it is not an added position vector.

The same matrix E returns at the end to score output tokens.

L19’s block, with Qwen’s real component names



The abstraction is unchanged; these names specify Qwen’s implementation.

Grouped-query attention: 16 Q heads share 8 K/V heads

Why share? Keep many query views, but store half as many K/V heads in the cache.

The real model repeats this pattern: 16 query heads grouped over 8 key/value heads.

The same residual stream passes through 28 blocks

Every block adds an update; the residual stream carries the evolving $[T, 1024]$ representation.

Inspect one measured attention pattern

REAL MODEL DATA

Pick a selected layer/head pattern, then choose a query token and read its exact causal row.

selected: it → robot | selected: previous-token pattern | selected: broad-context pattern

layer 11, head 1 · selected by: largest measured attention from query 'it' to a content token

The | ·robot | ·picked | ·up | ·the | ·cup | ·because | ·it | ·was | ·empty

query: ·it attends to…

·robot	0.75
·cup	0.10
·picked	0.04
The	0.03
·up	0.03
·it	0.02

Every row sums to approximately 1 and has zero future mass. Selected patterns are curated measurements, not causal explanations.

Attention is a measurement—not an explanation

one head

among 16 heads × 28 layers

curated

selected because its pattern was interpretable

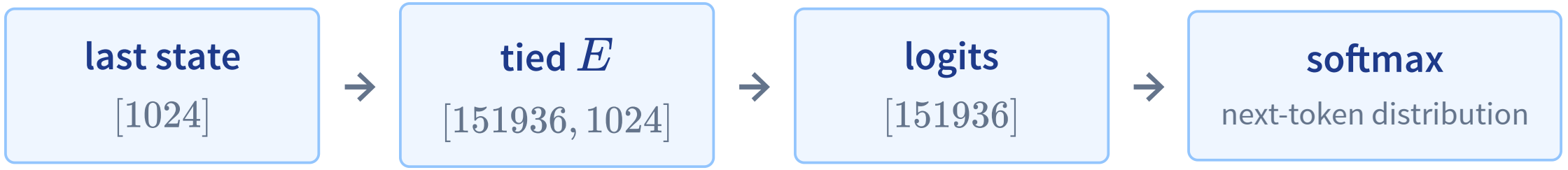
not causal proof

large weight does not prove decisive influence

Use attention to inspect a computation—not to claim the model's full reason.

Jain & Wallace, "Attention is not Explanation," NAACL 2019.

The final state is scored by the tied embedding matrix



Gradient	0.985
The	0.004
A	0.004
all other tokens	~0.007

Real Qwen trace after the non-thinking template for “Explain gradient descent in one sentence.”

Append one token, then score again

REAL MODEL DATA

Step through an exact Qwen continuation; every distribution is conditioned on the enlarged context.

Qwen3-0.6B post-trained assistant · non-thinking prompt · greedy trace.

next distribution: step 1 of 8


```
user: Explain gradient descent in one
sentence.
assistant:
```

append next token

reset

Candidate labels use · for a leading space. Every click reads a newly conditioned distribution.

Inference has two phases

1. prefill

prompt tokens in parallel

Build hidden states and K/V cache for the whole prompt.

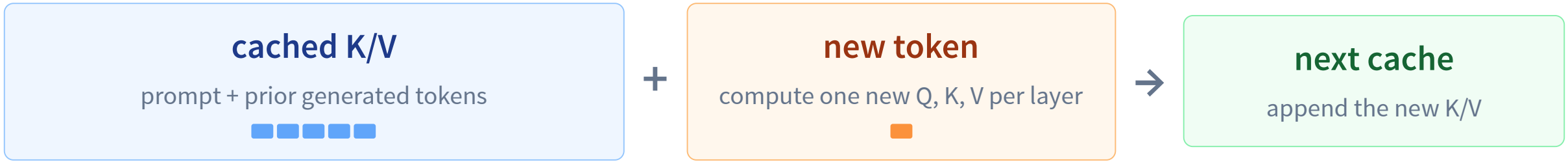
2. decode

one new token

Generate sequentially, one distribution per step.

Parallel prefill; sequential decode.

KV cache reuses the past at every layer



The new query still attends over every cached source;
caching avoids recomputing old K/V and hidden states.

How much recomputation does the cache avoid? LIVE

Step through prefill and decode; compare token-work with and without cached K/V.

Prefill processes all four prompt tokens and builds four K/V slots.

`The` `robot` `picked` `up` `the`

cached K/V slots

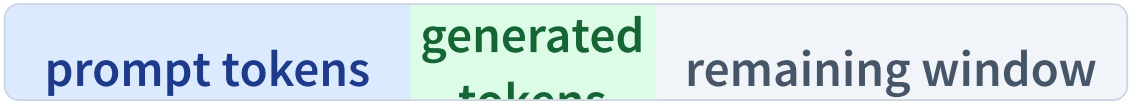
4 token positions computed without cache

4 token positions computed with cache

decode next token | reset

For three output tokens: without cache 4 + 5 + 6 = 15 token positions; with cache 4 + 1 + 1 = 6.

Context is working memory—and cache memory



window

prompt + output share Qwen’s 32k context

cache memory

grows with layers × KV heads × length

latency

decode remains token-by-token

The cache buys speed by spending memory.

The complete Python inference path

```
# 1. load tokenizer + post-trained weights
from transformers import AutoModelForCausalLM, AutoTokenizer
tok = AutoTokenizer.from_pretrained("Qwen/Qwen3-0.6B")
model = AutoModelForCausalLM.from_pretrained(
   "Qwen/Qwen3-0.6B", torch_dtype="auto", device_map="auto")
# 2. serialize the role/content messages
text = tok.apply_chat_template(messages, tokenize=False,
   add_generation_prompt=True, enable_thinking=False)
# 3. tokenize and move tensors beside the model
inputs = tok(text, return_tensors="pt").to(model.device)
# 4. autoregressively append new token IDs
all_ids = model.generate(**inputs, max_new_tokens=64,
   do_sample=True, temperature=.7, top_p=.8, top_k=20)
# 5. remove the prompt IDs, then decode only the answer
new_ids = all_ids[0, inputs.input_ids.shape[1]:]
answer = tok.decode(new_ids, skip_special_tokens=True)
```

The same path can run in the browser

```
import { pipeline } from "@huggingface/transformers";

const qwen = await pipeline(
  "text-generation",
  "onnx-community/Qwen3-0.6B-ONNX",
  { device: "webgpu", dtype: "q4f16" },
);

const messages = [{
  role: "user",
  content: "Explain gradient descent.",
}];
const result = await qwen(messages, {
  max_new_tokens: 64,
  do_sample: true, temperature: 0.7, top_p: 0.8, top_k: 20,
});
```

instant path

audited traces embedded in the deck

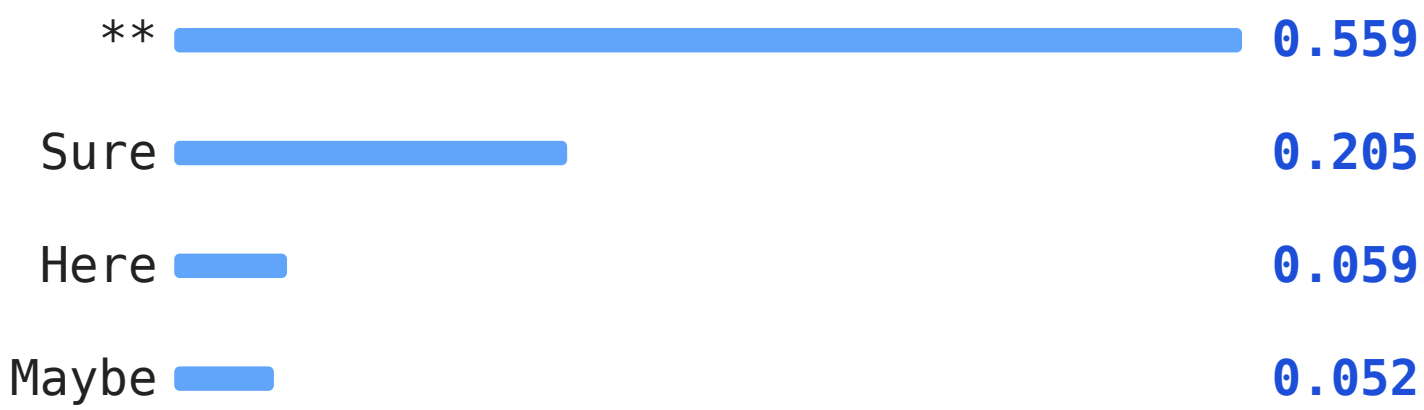
optional live path

several-hundred-MB download + compatible WebGPU

The concepts and default demos never depend on the live download succeeding.

One distribution, two selection rules

Qwen context: `Write a creative name for a friendly blue robot.`



greedy

always choose argmax: **

sampling

draw according to probability: varied path

The weights and distribution are fixed; only the selection rule changes.

Temperature reshapes; top-k and top-p trim

temperature	: sharper : unchanged : flatter	Ranking stays the same.
top-k = 3	keep **, Sure, Here	Remove every token below rank 3.
top-p = .8	.559 + .205 + .059 = .823	Keep the smallest prefix reaching 0.8.
renormalize	sample only from the kept candidates	

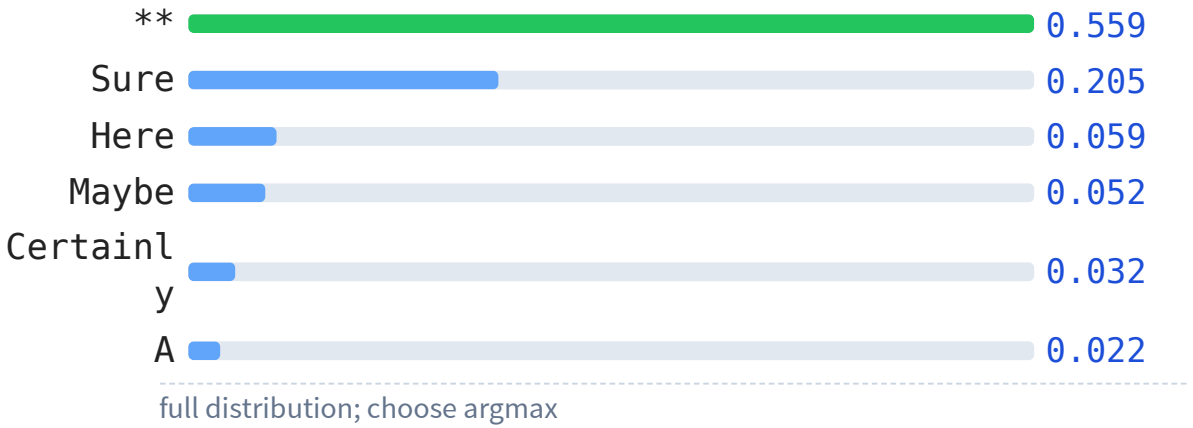
Run Qwen with honest decoding presets

REAL MODEL DATA

Every preset has its own exact distributions and continuation;
append repeatedly or optionally run the same settings live.

greedy | recommended non-thinking | recommended thinking

argmax | thinking OFF




```
user: Write a creative name for a friendly blue robot.
assistant:
```

append next token | reset | optional: run live

Precomputed controls are exact. Candidate labels use · for a leading space; thinking traces begin with <think>.

Prompting changes tokens in context—not weights

role	task	audience	examples	format
You are a patient tutor.	Explain gradient descent.	Use language for a 10-year-old.	Input/output pairs specify a pattern.	One sentence; no equations.

zero-shot + examples = few-shot prompt

general prompt	“Gradient descent is a method used to minimize a function …”
10-year-old prompt	“Gradient descent is like a way to get closer to the best answer …”

Real greedy Qwen outputs. More instructions and examples are simply more conditioning tokens.

Thinking mode changes the generation protocol

thinking OFF

```
<think>

</think>
```

Template pre-fills an empty block, then Qwen answers.

$$T = .7,\ p = .8,\ k = 20$$

thinking ON

```
<|im_start|>assistant
…
```

Qwen generates **<think>…</think>** before its answer.

$$T = .6,\ p = .95,\ k = 20$$

Reasoning text can help on hard tasks, but costs tokens and is not guaranteed correct or causally faithful.

A fluent answer can still be fabricated

user Who won the 2031 Turing Award?

real Qwen output “As of the latest information available (2024), the 2031 Turing Award was awarded to …”

correct behavior 2031 is in the future; abstain or retrieve current evidence.

changed today	**did not change**
template, context, decoding	weights, knowledge, available actions

L22: choose prompting vs RAG/tools vs SFT/LoRA—and see Qwen become a frozen interpreter for PAW.